

BOOK 9

PARTICLE PHYSICS

11 Particle physics

11.1 Atoms, nuclei and radiation

Candidates should be able to:

- 1 infer from the results of the α -particle scattering experiment the existence and small size of the nucleus
- 2 describe a simple model for the nuclear atom to include protons, neutrons and orbital electrons
- 3 distinguish between nucleon number and proton number
- 4 understand that isotopes are forms of the same element with different numbers of neutrons in their nuclei
- 5 understand and use the notation A_ZX for the representation of nuclides
- 6 understand that nucleon number and charge are conserved in nuclear processes
- 7 describe the composition, mass and charge of α -, β - and γ -radiations (both β^- (electrons) and β^+ (positrons) are included)
- 8 understand that an antiparticle has the same mass but opposite charge to the corresponding particle, and that a positron is the antiparticle of an electron
- 9 state that (electron) antineutrinos are produced during β^- decay and (electron) neutrinos are produced during β^+ decay
- 10 understand that α -particles have discrete energies but that β -particles have a continuous range of energies because (anti)neutrinos are emitted in β -decay
- 11 represent α - and β -decay by a radioactive decay equation of the form ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^4_2\alpha$
- 12 use the unified atomic mass unit (u) as a unit of mass

11.2 Fundamental particles

Candidates should be able to:

- 1 understand that a quark is a fundamental particle and that there are six flavours (types) of quark: up, down, strange, charm, top and bottom
- 2 recall and use the charge of each flavour of quark and understand that its respective antiquark has the opposite charge (no knowledge of any other properties of quarks is required)
- 3 recall that protons and neutrons are not fundamental particles and describe protons and neutrons in terms of their quark composition
- 4 understand that a hadron may be either a baryon (consisting of three quarks) or a meson (consisting of one quark and one antiquark)
- 5 describe the changes to quark composition that take place during β^- and β^+ decay
- 6 recall that electrons and neutrinos are fundamental particles called leptons

- 1 The spontaneous and random decay of a radioactive substance involves the emission of either α -radiation or β -radiation and/or γ -radiation.

(a) Explain what is meant by *spontaneous* decay.

.....
.....
..... [2]

(b) State the type of emission, one in each case, that

(i) is not affected by electric and magnetic fields,

..... [1]

(ii) produces the greatest density of ionisation in a medium,

..... [1]

(iii) does not directly result in a change in the proton number of the nucleus,

..... [1]

(iv) has a range of energies, rather than discrete values.

..... [1]

- 2 An α -particle A approaches and passes by a stationary gold nucleus N. The path is illustrated in Fig. 7.1.

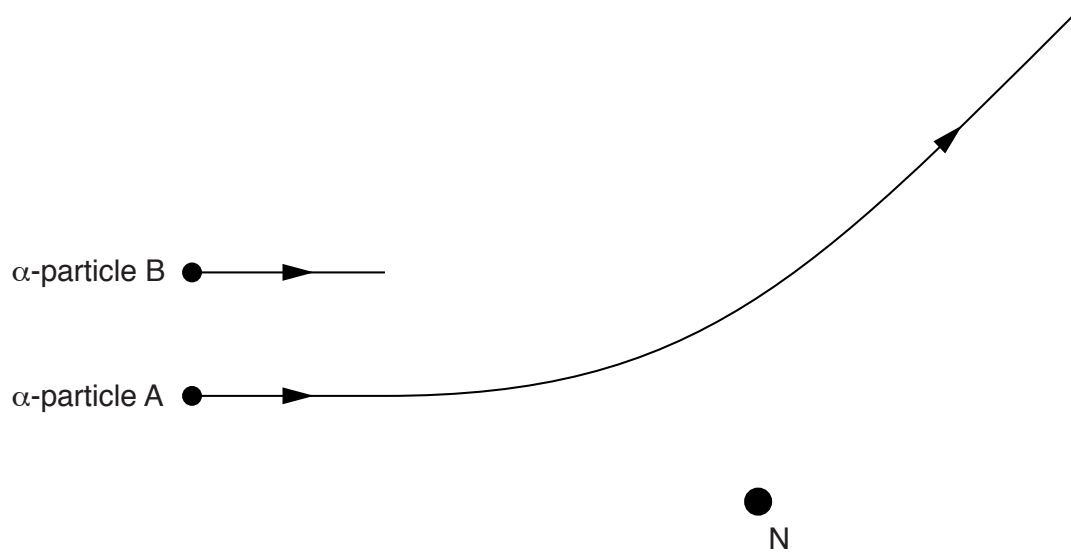


Fig. 7.1

- (a) On Fig. 7.1, mark the angle of deviation D of this α -particle as a result of passing the nucleus N. [1]
- (b) A second α -particle B has the same initial direction and energy as α -particle A. On Fig. 7.1, complete the path of α -particle B as it approaches and passes by the nucleus N. [2]
- (c) State what can be inferred about atoms from the observation that very few α -particles experience large deviations.

.....

.....

..... [2]

- (d) The nucleus N could be one of several different isotopes of gold.

Suggest, with an explanation, whether different isotopes of gold would give rise to different deviations of a particular α -particle.

.....

.....

..... [2]

- 3 Tungsten-184 ($^{184}_{74}\text{W}$) and tungsten-185 ($^{185}_{74}\text{W}$) are two isotopes of tungsten.

Tungsten-184 is stable but tungsten-185 undergoes β -decay to form rhenium (Re).

- (a) Explain what is meant by *isotopes*.

.....

 [2]

- (b) The β -decay of nuclei of tungsten-185 is spontaneous and random.

State what is meant by

- (i) *spontaneous* decay,

.....
 [1]

- (ii) *random* decay.

.....
 [1]

- (c) Complete the nuclear equation for the β -decay of a tungsten-185 nucleus.



4 One of the isotopes of uranium is uranium-238 (${}^{238}_{92}\text{U}$).

9702/21/M/J/10/Q7

(a) State what is meant by *isotopes*.

.....
..... [2]

(b) For a nucleus of uranium-238, state

(i) the number of protons,

number = [1]

(ii) the number of neutrons.

number = [1]

(c) A uranium-238 nucleus has a radius of $8.9 \times 10^{-15} \text{ m}$.

Calculate, for a uranium-238 nucleus,

(i) its mass,

mass = kg [2]

(ii) its mean density.

density = kg m^{-3} [2]

(d) The density of a lump of uranium is $1.9 \times 10^4 \text{ kg m}^{-3}$.

Using your answer to (c)(ii), suggest what can be inferred about the structure of the atom.

.....
.....
..... [2]

- 5 (a) The radioactive decay of some nuclei gives rise to the emission of α -particles.
State

(i) what is meant by an α -particle,

.....[1]

(ii) two properties of α -particles.

1.

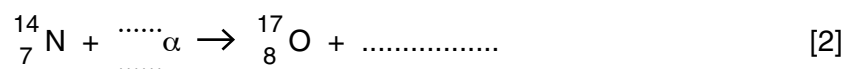
.....

2.

.....[2]

- (b) One possible nuclear reaction involves the bombardment of a stationary nitrogen-14 nucleus by an α -particle to form oxygen-17 and another particle.

(i) Complete the nuclear equation for this reaction.



- (ii) The total mass-energy of the nitrogen-14 nucleus and the α -particle is less than that of the particles resulting from the reaction. This mass-energy difference is 1.1 MeV.

1. Suggest how it is possible for mass-energy to be conserved in this reaction.

.....

.....[1]

2. Calculate the speed of an α -particle having kinetic energy of 1.1 MeV.

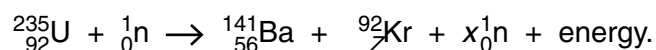
speed = m s⁻¹ [4]

- 6 (a) Uranium (U) has at least fourteen isotopes.
Explain what is meant by *isotopes*.

.....

 [2]

- (b) One possible nuclear reaction involving uranium is



- (i) State three quantities that are conserved in a nuclear reaction.

1.

 2.

 3.
 [3]

- (ii) For this reaction, determine the value of

1. Z ,

$Z =$ [1]

2. x .

$x =$ [1]

- 7 The results of the α -particle scattering experiment provided evidence for the existence and small size of the nucleus.

(a) State the result that provided evidence for

- (i) the small size of the nucleus, compared with the atom,

.....
.....
.....[2]

- (ii) the nucleus being charged and containing the majority of the mass of the atom.

.....
.....
.....[2]

(b) The α -particles in this experiment originated from the decay of a radioactive nuclide.

Suggest two reasons why β -particles from a radioactive source would be inappropriate for this type of scattering experiment.

1.
.....
2.
.....
.....[2]

Do not allow β -particles have negative charge or β -particles have high speed

- 8 (a) Two isotopes of the element uranium are $^{235}_{92}\text{U}$ and $^{238}_{92}\text{U}$.

Explain the term *isotope*.

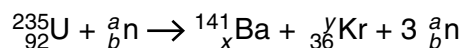
.....

 [2]

- (b) (i) In a nuclear reaction, proton number and neutron number are conserved. Other than proton number and neutron number, state a quantity that is conserved in a nuclear reaction.

..... [1]

- (ii) When a nucleus of uranium-235 absorbs a neutron, the following reaction may take place.



State the values of a , b , x and y .

$a =$

$b =$

$x =$

$y =$

[3]

- (c) When the nucleus of $^{238}_{92}\text{U}$ absorbs a neutron, the nucleus decays, emitting an α -particle. State the proton number and nucleon number of the nucleus that is formed as a result of the emission of the α -particle.

proton number =

nucleon number =

[2]

- 9 (a) State the experimental observations that show radioactive decay is

- (i) spontaneous,

..... [1]

- (ii) random.

..... [1]

- (b) On Fig. 7.1, complete the charge and mass of α -particles, β -particles and γ -radiation. Give example speeds of α -particles and γ -radiation emitted by a laboratory source.

	α -particle	β -particle	γ -radiation
charge			0
mass	4u		
speed		up to $0.99c$	

Fig. 7.1

[3]

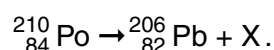
- (c) Explain the process by which α -particles lose energy when they pass through air.

.....

 [2]

9702/21/M/J/12/Q7

- 10 (a) The spontaneous decay of polonium is shown by the nuclear equation



- (i) State the composition of the nucleus of X.

.....
 [1]

- (ii) The nuclei X are emitted as radiation. State two properties of this radiation.

1.

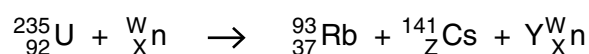
 2.
 [2]

- (b) The mass of the polonium (Po) nucleus is greater than the combined mass of the nuclei of lead (Pb) and X. Use a conservation law to explain qualitatively how this decay is possible.

.....

 [3]

- 11 (a) A nuclear reaction occurs when a uranium-235 nucleus absorbs a neutron. The reaction may be represented by the equation:



State the number represented by the letter

W Y

X Z [3]

- (b) The sum of the masses on the left-hand side of the equation in (a) is not the same as the sum of the masses on the right-hand side.

Explain why mass seems not to be conserved.

.....

..... [2]

- 12 A radioactive source emits α -radiation and γ -radiation.

Explain how it may be shown that the source does not emit β -radiation using

- (a) the absorption properties of the radiation,

.....

.....

.....

.....

..... [2]

- (b) the effects of a magnetic field on the radiation.

.....

.....

.....

.....

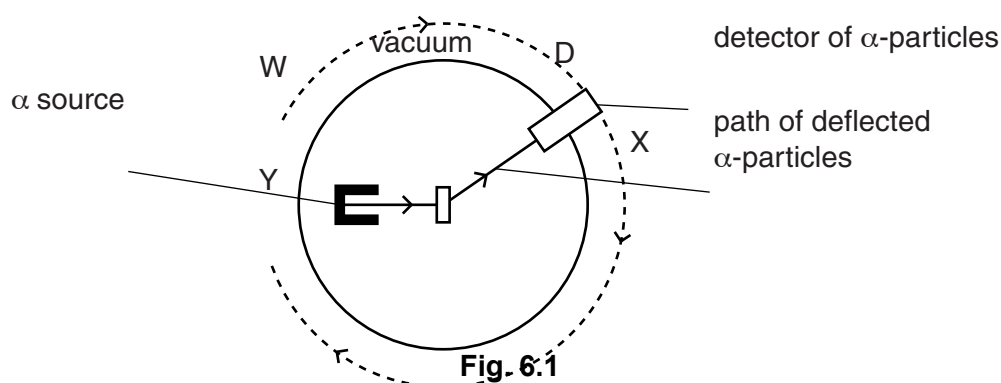
..... [2]

13 (a) Describe the structure of an atom of the nuclide $^{235}_{92}\text{U}$.

9702/21/O/N/12/Q6

.....
 [2]

(b) The deflection of α -particles by a thin metal foil is investigated with the arrangement shown in Fig. 6.1. All the apparatus is enclosed in a vacuum.



The detector of α -particles, D, is moved around the path labelled WXY.

(i) Explain why the apparatus is enclosed in a vacuum.

.....
 [1]

(ii) State and explain the readings detected by D when it is moved along WXY.

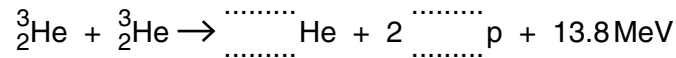
.....

 [3]

(c) A beam of α -particles produces a current of 1.5 pA. Calculate the number of α -particles per second passing a point in the beam.

number = s^{-1} [3]

- 14** A nuclear reaction between two helium nuclei produces a second isotope of helium, two protons and 13.8 MeV of energy. The reaction is represented by the following equation.



- (a)** Complete the nuclear equation. [2]

- (b)** By reference to this reaction, explain the meaning of the term *isotope*.

.....
 [2]

- (c)** State the quantities that are conserved in this nuclear reaction.

.....

 [2]

- (d)** Radiation is produced in this nuclear reaction.

State

- (i)** a possible type of radiation that may be produced,

..... [1]

- (ii)** why the energy of this radiation is less than the 13.8 MeV given in the equation.

..... [1]

- (e)** Calculate the minimum number of these reactions needed per second to produce power of 60 W.

number = s⁻¹ [2]

15 (a) β -radiation is emitted during the spontaneous radioactive decay of an unstable nucleus.

(i) State the nature of a β -particle.

..... [1]

(ii) State two properties of β -radiation.

1.

2. [2]

(iii) Explain the meaning of *spontaneous radioactive decay*.

..... [1]
.....

(b) The following equation represents the decay of a nucleus of hydrogen-3 by the emission of a β -particle.

Complete the equation.



(c) The β -particle is emitted with an energy of $5.7 \times 10^3 \text{ eV}$.

Calculate the speed of the β -particle.

speed = ms^{-1} [3]

(d) A different isotope of hydrogen is hydrogen-2 (deuterium). Describe the similarities and differences between the atoms of hydrogen-2 and hydrogen-3.

.....
.....
..... [2]

16 (a) Describe the two main results of the α -particle scattering experiment. 9702/21/M/J/13/Q7

result 1:

.....

result 2:

.....

[3]

(b) Relate each of the results in **(a)** with the conclusions that were made about the nature of atoms.

result 1:

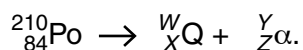
.....

result 2:

.....

[3]

17 A polonium nucleus $^{210}_{84}\text{Po}$ is radioactive and decays with the emission of an α -particle. The nuclear reaction for this decay is given by 9702/22/M/J/13/Q7



(a) (i) State the values of W

X

Y

Z

[2]

(ii) Explain why mass seems not to be conserved in the reaction.

.....

..... [2]

(b) The reaction is spontaneous. Explain the meaning of *spontaneous*.

.....

..... [1]

- 18 (a) An electric field is set up between two parallel metal plates in a vacuum. The deflection of α -particles as they pass between the plates is shown in Fig. 7.1.

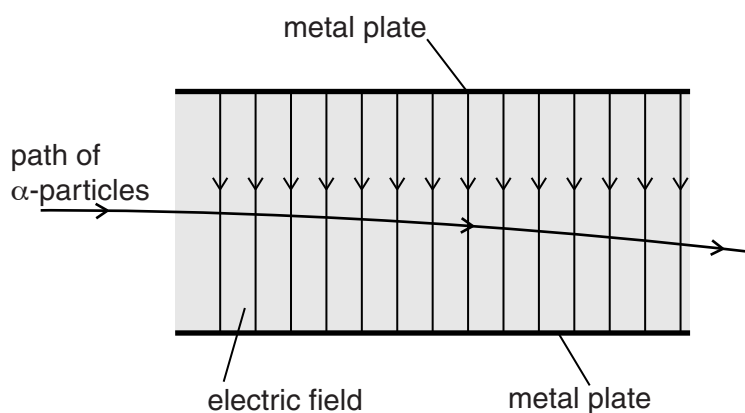


Fig. 7.1

The electric field strength between the plates is reduced. The α -particles are replaced by β -particles. The deflection of β -particles is shown in Fig. 7.2.

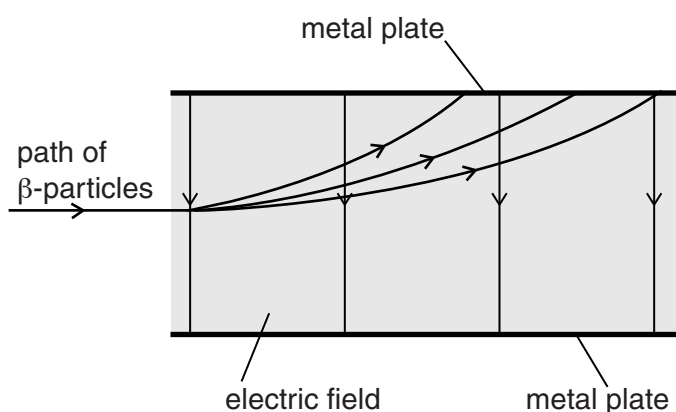


Fig. 7.2

- (i) State one similarity of the electric fields shown in Fig. 7.1 and Fig. 7.2.

.....
[1]

- (ii) The electric field strength in Fig. 7.2 is less than that in Fig. 7.1. State two methods of reducing this electric field strength.

1.
 2.

[2]

- (iii) By reference to the properties of α -particles and β -particles, suggest three reasons for the differences in the deflections shown in Fig. 7.1 and Fig. 7.2.

1.
-
2.
-
3.
-

[3]

- (b) A source of α -particles is uranium-238. The nuclear reaction for the emission of α -particles is represented by



State the values of W

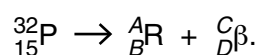
X

Y

Z

[2]

- (c) A source of β -particles is phosphorus-32. The nuclear reaction for the emission of β -particles is represented by



State the values of A

B

C

D

[1]

19 (a) State what is meant by

α -particle:

β -particle:

γ -radiation: [2]

(b) Describe the changes to the proton number and the nucleon number of a nucleus when emission occurs of

(i) an α -particle,

..... [1]

(ii) a β -particle,

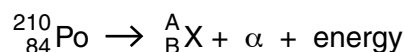
..... [1]

(iii) γ -radiation.

..... [1]

20 In the decay of a nucleus of $^{210}_{84}\text{Po}$, an α -particle is emitted with energy 5.3 MeV.

The emission is represented by the nuclear equation



(a) (i) On Fig. 7.1, complete the number and name of the particle, or particles, represented by A and B in the nuclear equation.

	number	name of particle or particles
A		
B		

Fig. 7.1

[1]

(ii) State the form of energy given to the α -particle in the decay of $^{210}_{84}\text{Po}$.

..... [1]

- (b) A sample of polonium $^{210}_{84}\text{Po}$ emits 7.1×10^{18} α -particles in one day.
 Calculate the mean power output from the energy of the α -particles.

power = W [2]

- 9702/22/M/J/15/Q7

21 A uranium-235 nucleus absorbs a neutron and then splits into two nuclei. A possible nuclear reaction is given by



- (a) State the constituent particles of the uranium-235 nucleus.
 [1]
- (b) Complete Fig. 7.1 for this reaction.

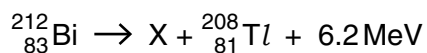
	value
a	
b	
c	
d	

Fig. 7.1

- (c) Suggest a possible form of energy released in this reaction.
 [1]
- (d) Explain, using the law of mass-energy conservation, how energy is released in this reaction.

 [2]

22 The equation represents the spontaneous radioactive decay of a nucleus of bismuth-212.



- (a) (i)** Explain the meaning of *spontaneous* radioactive decay.

.....
[1]

- (ii)** State the constituent particles of X.

.....[1]

- (b) (i)** Use the conservation of mass-energy to explain the release of 6.2MeV of energy in this reaction.

.....

[2]

- (ii)** Calculate the energy, in joules, released in this reaction.

energy = J [1]

23 A neutron decays by emitting a β^- particle.

9702/22/F/M/16/Q6

(a) Complete the equation below for this decay.

$${}^1_0\text{n} \rightarrow \begin{array}{c} \text{.....} \\ \text{.....} \end{array} + \begin{array}{c} \text{.....} \\ \text{.....} \end{array} \beta^- + \begin{array}{c} \text{.....} \\ \text{.....} \end{array} \bar{\nu} \quad [2]$$

(b) State the name of the particle represented by the symbol $\bar{\nu}$.

..... [1]

(c) State the name of the class (group) of particles that includes β^- and $\bar{\nu}$.

..... [1]

(d) State

(i) the quark structure of the neutron,

..... [1]

(ii) the change to the quark structure when the neutron decays.

.....

..... [1]

24 (a) Give one example of

a hadron:

a lepton: [1]

(b) Describe, in terms of the simple quark model,

(i) a proton,

.....[1]

(ii) a neutron.

.....[1]

(c) Beta particles may be emitted during the decay of an unstable nucleus of an atom. The emission of a beta particle is due to the decay of a neutron.

(i) Complete the following word equation for the particles produced in this reaction.

neutron $\rightarrow$ + + [1]

(ii) State the change in quark composition of the particles during this reaction.

.....[1]

25 (a) State the name of the class (group) to which each of the following belongs:

electron neutrino

neutron proton [2]

(b) A proton may decay into a neutron together with two other particles.

(i) Complete the following to give an equation that represents this proton decay.

${}^1_1\text{p} \rightarrow \dots\dots\dots \text{n} + \dots\dots\dots + \dots\dots\dots$ [2]

(ii) Write an equation for this decay in terms of quark composition.

(iii) State the name of the force responsible for this decay. [1]

.....[1]

26 (a) Distinguish between an α -particle and a β^+ -particle.

9702/23/M/J/16/Q8

.....

.....

.....

.....

.....[3]

(b) State the equation that shows the decay of a particle in a nucleus that results in β^+ emission. All particles in the equation should be shown in the notation that is usually used for the representation of nuclides.

[2]

(c) (i) State the quark composition of

1. a proton,

.....

2. a neutron.

.....

[2]

(ii) Use the quark model to explain the charge on a proton.

.....

.....

.....[1]

27 (a) State **one** difference between a hadron and a lepton.

.....
[1]

(b) (i) State the quark composition of a proton and of a neutron.

proton:
 neutron:
[2]

(ii) Use your answer in (i) to determine the quark composition of an α -particle.

quark composition:[1]

(c) The results of the α -particle scattering experiment provide evidence for the structure of the atom.

result 1: The vast majority of α -particles pass straight through the metal foil or are deviated by small angles.

result 2: A very small minority of α -particles are scattered through angles greater than 90° .

State what may be inferred from

(i) result 1,

.....
[1]

(ii) result 2.

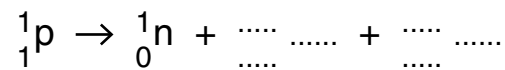
.....

[2]

28 (a) State **one** difference between a hadron and a lepton.

.....
[1]

(b) A proton within a nucleus decays to form a neutron and two other particles. A partial equation to represent this decay is



(i) Complete the equation. [2]

(ii) State the name of the interaction or force that gives rise to this decay.

.....[1]

(iii) State three quantities that are conserved in the decay.

1.

2.

3.

[3]

(c) Use the quark composition of a proton to show that it has a charge of $+e$, where e is the elementary charge.

Explain your working.

[3]

29 (a) Use the quark model to show that

9702/21/M/J/17/Q7

(i) the charge on a proton is $+e$,

.....[1]

(ii) the charge on a neutron is zero.

.....[1]

(b) A nucleus of $^{90}_{38}\text{Sr}$ decays by the emission of a β^- particle. A nucleus of $^{64}_{29}\text{Cu}$ decays by the emission of a β^+ particle.

(i) In Fig. 7.1, state the nucleon number and proton number for the nucleus produced in each of these decay processes.

	nucleus formed by β^- decay	nucleus formed by β^+ decay
nucleon number		
proton number		

Fig. 7.1

[1]

(ii) State the name of the force responsible for β decay.

.....[1]

(iii) State the names of the leptons produced in each of the decay processes.

β^- decay:

β^+ decay:[1]

9702/22/M/J/17/Q8

30 (a) Describe **two** differences between the decay of a nucleus that emits a β^- particle and the decay of a nucleus that emits a β^+ particle.

1.

.....

2.

.....

[2]

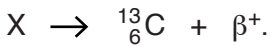
(b) In a simple quark model there are three types of quark. State the composition of the proton and of the neutron in terms of these three quarks.

proton:

neutron:

[1]

- 31** A stationary nucleus X decays by emitting a β^+ particle to form a nucleus of carbon-13 ($^{13}_6\text{C}$). An incomplete equation to represent this decay is



- (a)** State the name of the class (group) of particles that includes β^+ .
[1]
- (b)** For nucleus X, state the number of
 protons,
 neutrons.[1]
- (c)** The carbon-13 nucleus has a mass of $2.2 \times 10^{-26}\text{ kg}$. Its kinetic energy as a result of the decay process is 0.80 MeV .
 Calculate the speed of this nucleus.

speed = ms^{-1} [3]

- (d)** Explain why the sum of the kinetic energies of the carbon-13 nucleus and the β^+ particle cannot be equal to the total energy released by the decay process.

[1]
-

- 32 (a) A nucleus X decays by emitting a β^+ particle to form a new nucleus, ${}^{23}_{11}\text{Na}$.

State the number of nucleons and the number of neutrons in nucleus X.

number of nucleons =

number of neutrons =

[2]

- (b) State one similarity and one difference between a β^+ particle and a β^- particle.

similarity:

difference:

[2]

- 33 A β^- particle from a radioactive source is travelling in a vacuum with kinetic energy 460 eV. The particle enters a uniform electric field at a right-angle and follows the path shown in Fig. 7.1.

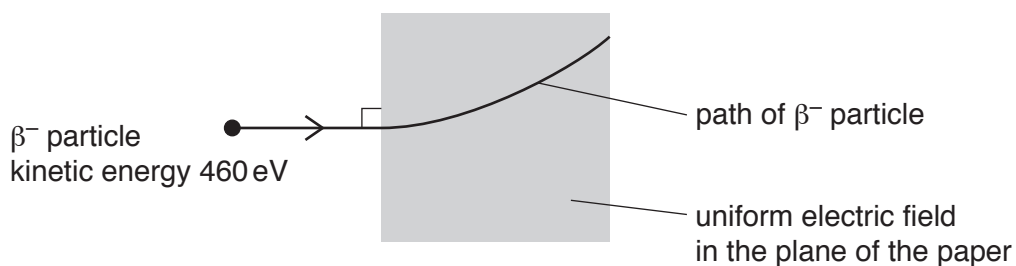


Fig. 7.1

- (a) The direction of the electric field is in the plane of the paper.
On Fig. 7.1, draw an arrow to show the direction of the electric field.

[1]

- (b) Calculate the speed of the β^- particle before it enters the electric field.

speed = ms^{-1} [3]

- (c) Other β^- particles from the same radioactive source travel outside the electric field along the same incident path as that shown in Fig. 7.1.

State and briefly explain whether those β^- particles will all follow the same path inside the electric field.

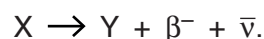
.....

.....

.....

.....[2]

- 34** A stationary nucleus X decays to form nucleus Y, as shown by the equation **9702/22/M/J/18/Q7**



- (a) In the above equation, draw a circle around all symbols that represent a lepton. [1]

- (b) State the name of the particle represented by the symbol $\bar{\nu}$.

.....[1]

- (c) Energy is released during the decay process. State the form of the energy that is gained by nucleus Y.

.....[1]

- (d) By comparing the compositions of X and Y, state and explain whether they are isotopes.

.....

.....

.....[2]

- (e) The quark composition of one nucleon in X is changed during the emission of a β^- particle. Describe this change to the quark composition.

.....

.....[1]

35 A graph of nucleon number A against proton number Z is shown in Fig. 7.1.

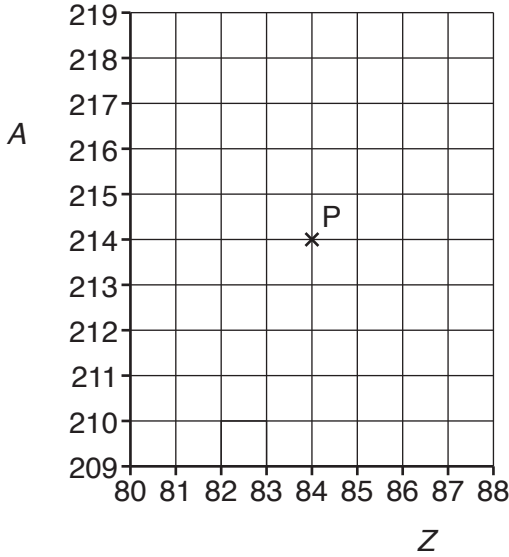


Fig. 7.1

The graph shows a cross (labelled P) that represents a nucleus P.

Nucleus P decays by emitting an α particle to form a nucleus Q.
Nucleus Q then decays by emitting a β^- particle to form a nucleus R.

- (a) On Fig. 7.1, use a cross to represent
- (i) nucleus Q (label this cross Q), [1]
 - (ii) nucleus R (label this cross R). [1]
- (b) State the name of the class (group) of particles that includes the β^- particle.
-[1]
- (c) The quark composition of one nucleon in Q is changed during the emission of the β^- particle. Describe this change to the quark composition.
-
-[1]
-

- 36 (a) State what is meant by an *electric field*.

.....
 [1]

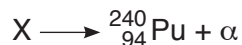
- (b) A particle of mass m and charge q is in a uniform electric field of strength E . The particle has acceleration a due to the field.

Show that

$$a = \frac{Eq}{m}.$$

[2]

- (c) A stationary nucleus X decays by emitting an α -particle to form a nucleus of plutonium, ${}^{240}_{94}\text{Pu}$, as shown.



- (i) Determine the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons =

[2]

- (ii) The total mass of the plutonium nucleus and the α -particle is less than that of nucleus X. Explain this difference in mass.

.....

 [2]

- (iii) The plutonium nucleus and the α -particle are both accelerated by the same uniform electric field.

Use the expression in (b) to determine the ratio

$$\frac{\text{acceleration of the } \alpha\text{-particle}}{\text{acceleration of the plutonium nucleus}} .$$

ratio = [2]

- 37 (a) In the following list, underline all particles that are leptons.

9702/22/O/N/18/Q8

antineutrino positron proton quark [1]

- (b) A stationary nucleus of magnesium-27, ${}^{27}_{12}\text{Mg}$, decays by emitting a β^- particle and γ radiation. An incomplete equation to represent this decay is



- (i) State the nucleon number and the proton number of nucleus X.

nucleon number =

proton number = [2]

- (ii) State the name of the interaction that gives rise to this decay.

.....[1]

- (iii) State **two** possible reasons why the sum of the kinetic energy of the β^- particle and the energy of the γ radiation is less than the total energy released during the decay of the magnesium nucleus.

1.
.....

2.
.....

[2]

- 38** A particle of mass m and charge q is in a uniform electric field of strength E . The particle has acceleration a due to the field.

(a) Show that

$$\frac{q}{m} = \frac{a}{E}.$$

[2]

- (b)** The particle has a charge of $4e$ where e is the elementary charge. The electric field strength is $3.5 \times 10^4 \text{ V m}^{-1}$. The acceleration of the particle is $1.5 \times 10^{12} \text{ m s}^{-2}$.

Use the expression in **(a)** to show that the mass of the particle is 9.0 u .

[2]

- (c)** The particle is a nucleus. State the number of protons and the number of neutrons in the nucleus.

number of protons =

number of neutrons =

[1]

- (d)** A second nucleus that is an isotope of the nucleus in **(c)** is in the same uniform electric field. State and explain whether the electric field produces, for the two nuclei, the same magnitudes of

(i) force,

.....

.....[1]

(ii) acceleration.

.....

.....[1]

- 39 (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .

State what may be inferred about the structure of the atom from this result.

.....

.....

.....

.....[2]

- (b) A hadron has an overall charge of $+e$, where e is the elementary charge. The hadron contains three quarks. One of the quarks is a strange (s) quark.

- (i) State the charge, in terms of e , of the strange (s) quark.

charge = [1]

- (ii) The other two quarks in the hadron have the same charge as each other.

By considering charge, determine a possible type (flavour) of the other two quarks.
Explain your working.

.....

.....[2]

- 40 (a) State what is meant by a *field line* (*line of force*) in an electric field.

.....
[1]

- (b) An electric field has two different regions X and Y. The field strength in X is less than that in Y. Describe a difference between the pattern of field lines (lines of force) in X and in Y.

.....
[1]

- (c) A particle P has a mass of 0.15 u and a charge of $-1e$, where e is the elementary charge.

- (i) Particle P and an α -particle are in the same uniform electric field. Calculate the ratio

$$\frac{\text{magnitude of acceleration of particle P}}{\text{magnitude of acceleration of } \alpha\text{-particle}}.$$

ratio = [3]

- (ii) Particle P is a hadron composed of only two quarks. One of them is a down (d) quark.

By considering charge, determine a possible type (flavour) of the other quark.
 Explain your working.

.....
[3]

- 41** A sample of a radioactive substance may decay by the emission of either α -radiation or β -radiation and/or γ -radiation.

State the type of radiation, one in each case, that:

- (a) consists of leptons

.....[1]

- (b) contains quarks

.....[1]

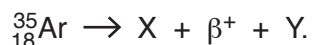
- (c) cannot be deflected by an electric field

.....[1]

- (d) has a continuous range of energies, rather than discrete values of energy.

.....[1]

- 42** (a) The decay of a nucleus ${}^{35}_{18}\text{Ar}$ by β^+ emission is represented by



A nucleus X and two particles, β^+ and Y, are produced by the decay.

State:

- (i) the proton number and the nucleon number of nucleus X

proton number =

nucleon number =

[1]

- (ii) the name of the particle represented by the symbol Y.

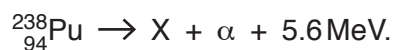
..... [1]

- (b) A hadron consists of two down quarks and one strange quark.

Determine, in terms of the elementary charge e , the charge of this hadron.

charge = [2]

- 43** A nucleus of plutonium-238 (${}^{238}_{94}\text{Pu}$) decays by emitting an α -particle to produce a new nucleus X and 5.6 MeV of energy. The decay is represented by



- (a)** Determine the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons = [2]

- (b)** Calculate the number of plutonium-238 nuclei that must decay in a time of 1.0 s to produce a power of 0.15 W.

number = [2]

- 44 A stationary nucleus of a radioactive isotope X decays by emitting an α -particle to produce a nucleus of neptunium-237 and 5.5 MeV of energy. The decay is represented by



- (a) Calculate the number of protons and the number of neutrons in a nucleus of X.

number of protons =

number of neutrons = [2]

- (b) Explain why the energy transferred to the α -particle as kinetic energy is less than the 5.5 MeV of energy released in the decay process.

.....
 [1]

- (c) A sample of X is used to produce a beam of α -particles in a vacuum. The number of α -particles passing a fixed point in the beam in a time of 30 s is 6.9×10^{11} .

- (i) Calculate the average current produced by the beam of α -particles.

current = A [2]

- (ii) Determine the total power, in W, that is produced by the decay of 6.9×10^{11} nuclei of X in a time of 30 s.

power = W [2]

45. 9702/22/F/M/21/Q7

- (a) The results of the α -particle scattering experiment provide evidence for the structure of the atom.

Result 1: The vast majority of the α -particles pass straight through the metal foil or are deviated by small angles.

Result 2: A very small minority of α -particles is scattered through angles greater than 90° .

State what may be inferred (deduced) from:

- (i) result 1

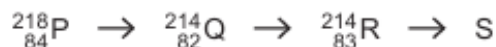
.....
 [1]

- (ii) result 2.

.....

 [2]

- (b) A radioactive decay sequence contains four nuclei, P, Q, R and S, as shown.



Nucleus S is an isotope of nucleus P.

- (i) Determine the proton number and the nucleon number of nucleus S.

proton number =
 nucleon number = [2]

- (ii) The quark composition of a nucleon in Q changes as Q decays to form R.

Describe this change to the quark composition of the nucleon.

.....
 [1]

[Total: 6]

46. 9702/22/M/J/21/Q6

- (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .

State what may be inferred about the structure of the atom from this result.

.....
.....
.....
..... [2]

- (b) An α -particle is made up of other particles. One of these particles is a proton.

State and explain whether a proton is a fundamental particle.

.....
..... [1]

- (c) A radioactive source produces a beam of α -particles in a vacuum. The average current produced by the beam is $6.9 \times 10^{-9} \text{ A}$.

Calculate the average number of α -particles passing a fixed point in the beam in a time of 1.0 minute.

number = [3]

47. 9702/23/M/J/21/Q6

(a) State the quark composition of:

(i) a proton

..... [1]

(ii) a neutron

..... [1]

(iii) an alpha-particle.

.....
..... [2]

(b) In the alpha-particle scattering experiment, alpha-particles were directed at a thin gold foil.

State what may be inferred from:

(i) the observation that most alpha-particles pass through the foil

..... [1]

(ii) the observation that some alpha-particles are scattered through angles greater than 90° .

.....
.....
..... [2]

48. 9702/21/O/N/21/Q6

- (a) Complete Table 6.1 to show the masses (in terms of the unified atomic mass unit u) and charges (in terms of the elementary charge e) of α , β^+ and β^- particles.

Table 6.1

	mass/ u	charge/ e
α -particle		
β^+ particle		
β^- particle		

[4]

- (b) Carbon-14 is radioactive and decays by emission of β^- particles.

- (i) Nuclei do not contain β^- particles.

Explain the origin of the β^- particle that is emitted from the nucleus during β^- decay.

.....

 [1]

- (ii) State the change in the quark composition of a carbon-14 nucleus when it emits a β^- particle.

..... [1]

- (iii) Suggest why the β^- particles are emitted with a range of different energies.

.....

 [2]

[Total: 8]

49. 9702/22/O/N/21/Q7

A stationary nucleus P of mass 243u decays by emitting an α -particle of mass 4u to form a different nucleus Q, as illustrated in Fig. 7.1.

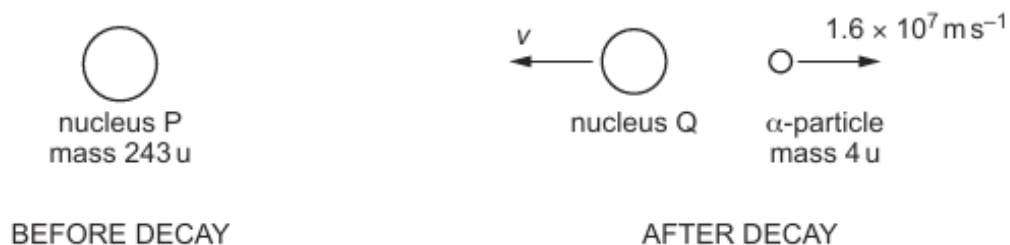


Fig. 7.1

The initial speed of the α -particle is $1.6 \times 10^7 \text{ m s}^{-1}$.

- (a) Use the principle of conservation of momentum to explain why the initial velocities of nucleus Q and the α -particle must be in opposite directions.

.....

 [2]

- (b) Determine the initial speed v of nucleus Q.

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

- (c) Calculate the initial kinetic energy, in MeV, of the α -particle.

kinetic energy = $\dots\dots\dots \text{ MeV}$ [3]

(d) A graph of number of neutrons N against proton number Z is shown in Fig. 7.2.

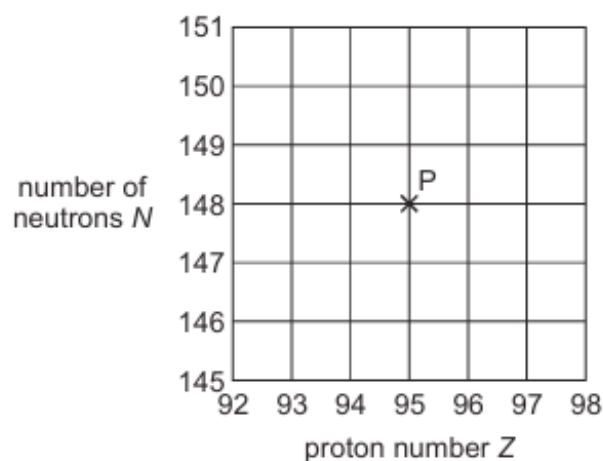


Fig. 7.2

The graph shows a cross that represents nucleus P.

A nucleus R has a nucleon number of 242 and is an isotope of nucleus P.

Nucleus R decays by emitting a β^- particle to form a different nucleus S.

(i) On Fig. 7.2, draw a cross to represent:

1. nucleus R (label this cross R)
2. nucleus S (label this cross S).

[2]

(ii) State the name of the other lepton, in addition to the β^- particle, that is emitted during the decay of nucleus R.

..... [1]

[Total: 10]

50. 9702/22/F/M/22/Q7

- (a) A nucleus of sodium-22 ($^{22}_{11}\text{Na}$) decays by emitting a β^+ particle. A different nucleus is formed by the decay.

- (i) State the name of another lepton that is produced by the decay.

..... [1]

- (ii) Determine the nucleon number and the proton number of the nucleus that is formed by the decay.

nucleon number =

proton number =

[2]

- (iii) The quark composition of a nucleon in the sodium-22 nucleus is changed during the decay.

Describe the change to the quark composition of the nucleon.

.....

..... [1]

- (b) A baryon consists of quarks that are the same flavour (type). The charge of the baryon is $-2e$, where e is the elementary charge.

- (i) Calculate, in terms of e , the charge of each quark.

charge = e [1]

- (ii) State a possible flavour (type) of the quarks.

..... [1]

[Total: 6]

51. 9702/21/M/J/22/Q7

(a) An unstable nucleus ${}^A_Z\text{X}$ decays by emitting a β^- particle.

(i) Determine quantitatively the changes, if any, in A and Z when X decays.

change in A =

change in Z =

[2]

(ii) In addition to the β^- particle, another lepton is emitted during the decay.

State the name of the other lepton that is emitted.

..... [1]

(b) A particle P is composed of an up quark (u) and a down antiquark ($\bar{d}$).

(i) Calculate the charge q of particle P in terms of e , where e is the elementary charge.

Show your working.

$q = \dots\dots\dots e$ [2]

(ii) Particle P belongs to **two** classes (groups) of particles.

State the names of these two classes.

1

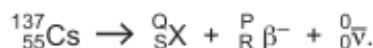
2

[2]

[Total: 7]

52. 9702/22/M/J/22/Q7

- (a) A nucleus of caesium-137 ($^{137}_{55}\text{Cs}$) decays by emitting a β^- particle to produce a nucleus of an element X and an antineutrino. The decay is represented by



- (i) State the number represented by each of the following letters.

P

Q

R

S

[2]

- (ii) State the name of the class (group) of particles that includes the β^- particle and the antineutrino.

..... [1]

- (b) A particle Y has a quark composition of ddd where d represents a down quark.

A particle Z has a quark composition of $\bar{u}d$ where $\bar{u}$ represents an up antiquark.

- (i) Show that the charges of particles Y and Z are equal.

[2]

- (ii) State and explain which particle is a meson and which particle is a baryon.

meson:

.....

baryon:

.....

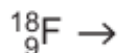
[2]

[Total: 7]

53. 9702/23/M/J/22/Q7

- (a) Fluorine-18 ($^{18}_9\text{F}$) is an isotope that decays to an isotope of oxygen (O) by the emission of a β^+ particle.

- (i) Complete the nuclear equation for the decay, including all the particles involved.



[3]

- (ii) A quark in the fluorine-18 nucleus changes flavour during the decay. State this change of flavour.

..... quark to quark.

[1]

- (b) A hadron has a charge of $-2e$, where e is the elementary charge.

- (i) State and explain whether the hadron is a meson or a baryon.

.....
.....
..... [2]

- (ii) State a possible quark composition for the hadron.

.....
..... [1]

[Total: 7]

54. 9702/21/O/N/22/Q6

- (a) A lepton is an example of a fundamental particle.

State what is meant by fundamental particle.

.....

..... [1]

- (b) A lambda particle Λ^0 is a hadron that consists of an up (u) quark, a down (d) quark and a strange (s) quark.

Show that the charge on the Λ^0 particle is zero.

[2]

- (c) The Λ^0 particle is unstable. It can decay into a neutron (n) and a pion (π^0) as shown by

$$\Lambda^0 \rightarrow n + \pi^0.$$

The π^0 particle consists of an up quark and an up antiquark.

- (i) Compare the properties of an up quark and an up antiquark.

.....

.....

..... [2]

- (ii) Explain why the neutron is classed as a baryon and the π^0 particle is classed as a meson.

.....

.....

..... [2]

[Total: 7]

55. 9702/22/O/N/22/Q7

- (a) Describe the structure of an **atom** of uranium-238, $^{238}_{92}\text{U}$.

.....

 [2]

- (b) The decay of uranium-238 is shown by the equation



For nucleus X, calculate the ratio, in C kg^{-1} , of

$$\frac{\text{charge}}{\text{mass}}.$$

ratio = C kg^{-1} [3]

- (c) Two particles P and Q each consist of three quarks. These quarks are up (u) or down (d) quarks.

Particle P has no overall charge.

Particle Q has an overall charge of $+2e$, where e is the elementary charge.

State the quark composition of:

- (i) particle P

..... [1]

- (ii) particle Q.

..... [1]

[Total: 7]

56. 9702/23/O/N/22/Q6

- (a) The nuclide $^{14}_6\text{C}$ (carbon-14) is unstable and undergoes β^- decay, emitting a high-energy electron and an antineutrino to form a new nuclide X. The equation for this decay is shown.



Complete the equation.

[2]

- (b) (i) State the equation for β^- decay in terms of the fundamental particles involved.

[1]

- (ii) Use your equation from (b)(i) to show how charge is conserved in β^- decay.

[1]

- (c) Neutrinos were first proposed to exist more than 20 years before they were directly detected, in order to explain a particular experimental observation about β -decay.

- (i) State an observation about β -decay that is explained by the existence of neutrinos.

.....

 [1]

- (ii) Suggest how the existence of neutrinos explains the observation in (c)(i).

.....

 [1]

[Total: 6]

57. 9702/22/F/M/23/Q7

- (a)** Nuclei X and Y are different isotopes of the same element.

Nucleus X is unstable and emits a β^+ particle to form nucleus Z.

By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus X is less than, the same as or greater than the charge of:

- (i)** nucleus Y

.....
..... [1]

- (ii)** nucleus Z.

.....
.....
..... [2]

- (b)** Hadrons can be divided into two groups (classes), P and Q. Group P is baryons.

- (i)** State the name of group Q.

..... [1]

- (ii)** Describe, in general terms, the quark structure of hadrons that belong to group Q.

.....
..... [1]

[Total: 5]

58. 9702/21/M/J/23/Q8

An isolated stationary nucleus X decays by emitting an α -particle to form a nucleus Y.

Nucleus Y and nucleus Z are isotopes of the same element.

- (a) By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus Y is less than, greater than or the same as the charge of:

- (i) nucleus Z

.....
..... [1]

- (ii) nucleus X.

.....
.....
..... [2]

- (b) Use the principle of conservation of momentum to explain why nucleus Y cannot be stationary immediately after the decay of nucleus X.

.....
.....
.....
..... [2]

[Total: 5]

59. 9702/22/M/J/23/Q8

- (a)** Nucleus P and nucleus Q are isotopes of the same element.

Nucleus Q is unstable and emits a β^- particle to form nucleus R.

- (i)** For nuclei P and Q, compare:

- the number of protons

.....

- the number of neutrons.

.....

[2]

- (ii)** When nucleus Q decays to form nucleus R, the quark composition of a nucleon changes.

State the change to the quark composition of the nucleon.

..... [1]

- (iii)** State the name of another particle that must be emitted from nucleus Q in addition to the β^- particle.

..... [1]

- (b)** A hadron consists of two charm quarks and one bottom quark.

Determine, in terms of the elementary charge e , the charge of the hadron.

charge = e [2]

[Total: 6]

60. 9702/23/M/J/23/Q7

- (a) Table 7.1 shows incomplete data for three flavours (types) of quark. The elementary charge is e .

Table 7.1

flavour	quark		antiquark	
	symbol	charge / e	symbol	charge / e
up	u	$+\frac{2}{3}$	$\bar{u}$	
down	d		$\bar{d}$	
charm	c		$\bar{c}$	

Complete Table 7.1 by inserting the missing charges. [2]

- (b) Using the symbols given in Table 7.1, state a possible quark combination for the following hadrons:

- (i) a neutral baryon

..... [1]

- (ii) a meson with a charge of $+e$.

..... [1]

- (c) Quarks are fundamental particles.

Electrons are in another group (class) of fundamental particle.

- (i) State the name of this group.

..... [1]

- (ii) State the name of another particle in this group.

..... [1]

[Total: 6]

61. 9702/21/O/N/23/Q7

- (a) The results of the α -particle scattering experiment led to the development of the nuclear model for the atom.

State the results that suggested that most of the mass of the atom is concentrated in a very small region and most of the atom is empty space.

.....

.....

.....

..... [2]

- (b) State the composition of γ -radiation.

..... [1]

- (c) Table 7.1 lists the names of three particles and possible classifications for them.

Table 7.1

particle name	classification		
	baryon	hadron	lepton
neutrino			
neutron			
positron			

Complete Table 7.1 by placing ticks (✓) in the boxes to indicate the classifications that apply to each particle. [2]

- (d) The discovery of a particle with an unusual charge was an important step in the development of the theory of quarks. The particle is a hadron with a mass of 2.19×10^{-27} kg and a charge of $+2e$, where e is the elementary charge.

- (i) Calculate the mass, in u, of the particle. Give your answer to three significant figures.

mass = u [1]

- (ii) Determine a possible quark composition of a hadron with a charge of $+2e$.
Explain your reasoning.

[2]

[Total: 8]

62. 9702/22/O/N/23/Q7

(a) In the following list, underline **all** the particles that are **not** fundamental.

antineutrino baryon nucleon positron [1]

(b) A nucleus of thorium-230 ($^{230}_{90}\text{Th}$) decays in stages, by emitting α -particles and β^- particles, to form a nucleus of lead-206 ($^{206}_{82}\text{Pb}$).

Determine the total number of α -particles and the total number of β^- particles that are emitted during the sequence of decays that form the nucleus of lead-206 from the nucleus of thorium-230.

number of α -particles =

number of β^- particles =

[2]

(c) A meson has a charge of $-1e$, where e is the elementary charge. The quark composition of the meson includes a charm antiquark.

State and explain a possible flavour (type) of the other quark in the meson.

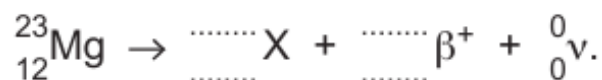
.....

..... [2]

[Total: 5]

63.9702/23/O/N/23/Q8

- (a) The nuclide ${}^{23}_{12}\text{Mg}$ is an isotope of magnesium that undergoes β^+ decay to form a new nuclide X according to the equation



Four numbers are missing from the equation.

- (i) For the nuclide ${}^{23}_{12}\text{Mg}$, state what is represented by the numbers 23 and 12.

23 represents:

12 represents: [2]

- (ii) Complete the equation by inserting the missing numbers. [2]

- (iii) State the name of the group (class) of fundamental particles to which the positron and neutrino belong.

..... [1]

- (b) A radioactive source emits particles from its nuclei when it decays.

Fig. 8.1 shows, for the source, the variation with kinetic energy of the number of particles emitted.

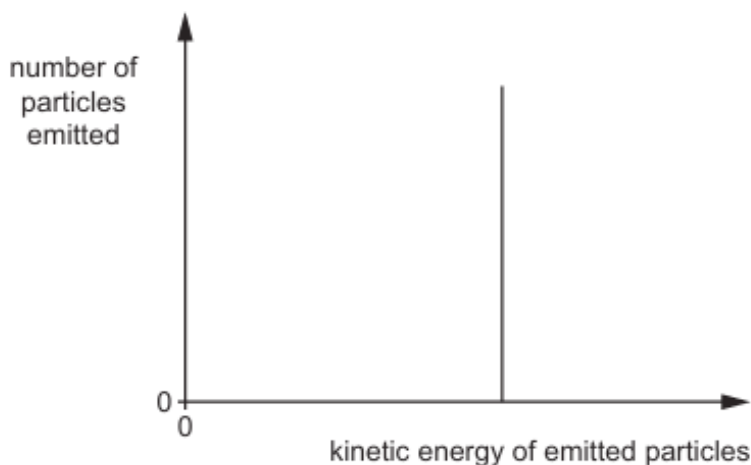


Fig. 8.1

State how Fig. 8.1 shows that these nuclei do **not** undergo beta-decay.

.....

..... [1]

[Total: 6]